

SMART CROP ADVISORY SYSTEM

Minor Project Synopsis

Submitted in partial fulfillment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY IN COMPUTER SCIENCE AND ENGINEERING

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CERTIFICATE

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This Minor Project is a record of the students’ original work carried out by them in partial fulfilment of the requirements for the award of the degree of Bachelor of Technology in Computer Science and Engineering from Dr. A.P.J. Abdul Kalam Institute of Technology, Tanakpur (Campus Institute of VMSBUTU, Dehradun), Uttarakhand, India.

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Figure 1: System Architecture Diagram for Smart Crop Advisory System (SCAS)

Figure 2: Workflow Diagram for Smart Crop Advisory System (SCAS)

Figure 3: ER Diagram for Smart Crop Advisory System (SCAS)

ABBREVIATIONS

- **AI** – Artificial Intelligence
- **ML** – Machine Learning
- **SCAS** – Smart Crop Advisory System
- **CNN** – Convolutional Neural Network
- **NLP** – Natural Language Processing
- **API** – Application Programming Interface
- **NPK** – Nitrogen, Phosphorus, Potassium
- **IoT** – Internet of Things
- **CRUD** – Create, Read, Update, Delete
- **UI** – User Interface
- **DB** – Database
- **SQL** – Structured Query Language
- **JPA** – Java Persistence API
- **REST** – Representational State Transfer
- **ERD** – Entity Relationship Diagram
- **WBS** – Work Breakdown Structure

ABSTRACT

Agriculture plays a crucial role in economic growth and food security, particularly in developing countries like India. However, farmers continue to face challenges such as unpredictable weather conditions, soil degradation, improper crop selection, inefficient fertilizer usage, and crop losses due to diseases and weeds. Traditional farming practices largely rely on experience or unverified local advice, which often leads to poor decision-making and reduced agricultural productivity. With the rapid advancement of Artificial Intelligence (AI) and Machine Learning (ML), agriculture is gradually shifting toward data-driven and precision-based practices.

The **Smart Crop Advisory System (SCAS)** is proposed as an intelligent, AI-driven web platform designed to assist farmers in making informed agricultural decisions. The system provides crop recommendations based on soil nutrients and weather conditions, optimized fertilizer suggestions, and early detection of crop diseases and weeds using deep learning techniques. Real-time weather alerts are integrated to help mitigate climate-related risks. Additionally, a community and blog module enables knowledge sharing among farmers. By integrating multiple advisory services into a unified platform, SCAS aims to enhance crop productivity, reduce losses, optimize resource usage, and promote sustainable farming practices through technology-driven solutions

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1. INTRODUCTION

Agriculture is the backbone of many economies, yet farmers continue to face challenges such as unpredictable weather, soil degradation, improper crop selection, and crop losses due to diseases and weeds. Traditional farming practices often rely on experience rather than data-driven insights, which increases production risks and reduces yield efficiency. With the advancement of Artificial Intelligence (AI) and Machine Learning (ML), modern agriculture is shifting toward precision-based decision making. The **Smart Crop Advisory System (SCAS)** is designed to assist farmers by providing intelligent recommendations for crop selection, fertilizer usage, and early detection of diseases and weeds. By analyzing soil nutrients, weather conditions, and crop images, the system delivers accurate, real-time advisories. SCAS aims to enhance productivity, minimize losses, and promote sustainable farming through technology-driven solutions.

1.1 Problem Statement

A majority of India's small and marginal farmers still depend on guesswork or unverified local advice from shopkeepers when making critical agricultural decisions regarding crop selection, fertilizer usage, and pest control.

This leads to:

- Ineffective farming practices,
- Excessive use of fertilizers and pesticides,
- Declining soil health, and
- Reduced agricultural productivity and income.

Without reliable data or expert guidance, farmers remain trapped in a cycle of low yields, high input costs, and environmental degradation.

1.2 Objectives

The main objectives of the proposed project are:

- To develop an AI-based crop recommendation system using soil and weather parameters.
- To design a fertilizer recommendation module for optimal fertilizer recommendation.
- To detect crop diseases and weeds using image-based deep learning models.
- To provide real-time weather alerts for risk mitigation.
- To enable farmers to make informed, data-driven, and sustainable farming decisions through a collaborative community platform.

1. Literature Review

Recent advancements in Artificial Intelligence (AI) and Machine Learning (ML) have significantly influenced modern agricultural practices by enabling data-driven decision-making. Several studies have focused on crop recommendation systems using soil parameters such as Nitrogen, Phosphorus, Potassium, pH, and weather conditions including temperature, humidity, and rainfall. Research findings indicate that machine learning algorithms, particularly Random Forest, achieve high accuracy in predicting suitable crops due to their ability to handle complex and non-linear agricultural data.

In the domain of fertilizer recommendation, existing systems utilize soil test reports and supervised learning models to suggest optimal fertilizer usage, thereby improving soil health and crop productivity. Similarly, crop disease detection has gained attention through the use of deep learning techniques, especially Convolutional Neural Networks (CNNs), which have demonstrated high accuracy in identifying plant diseases from leaf images.

While these studies report promising results, most existing solutions focus on individual components such as crop selection, fertilizer advisory, or disease detection in isolation. Limited work has been done on integrating these functionalities into a unified platform. Additionally, challenges related to real-world deployment, scalability, and accessibility for rural farmers remain inadequately addressed

2.1 Summary of Existing Solutions

- Most existing systems focus on single-function solutions, such as crop recommendation, fertilizer advisory, or disease detection independently.
- Machine learning models like Random Forest are widely used for crop and fertilizer recommendation due to their high accuracy on structured data.
- Deep learning models, especially CNNs, are commonly employed for crop disease detection using leaf images.
- Several solutions demonstrate high accuracy in controlled environments but lack real-world deployment readiness.
- Deployment strategies often rely on monolithic architectures, limiting scalability and flexibility.
- User accessibility is frequently overlooked, especially for farmers with low-end devices or poor internet connectivity.

2.2 Identification of Research Gaps

Despite significant advancements, the literature reveals several critical research gaps:

- **Lack of Integrated Systems:** Existing studies address crop recommendation, fertilizer optimization, and disease detection separately, with very few systems integrating all these components into a unified advisory platform.
- **Limited Real-World Deployment:** High accuracy results are often achieved using controlled datasets, which may not generalize well to diverse agro-climatic regions and real-field conditions.
- **Scalability and Architecture Issues:** Many solutions do not leverage cloud-native microservices, leading to poor scalability and difficulty in managing multiple AI models simultaneously.
- **Poor Accessibility for Rural Farmers:** Limited focus on offline functionality, low-bandwidth usage, and mobile-friendly platforms restrict adoption among small and marginal farmers.
- **Lack of Continuous Feedback and Community Support:** Most systems do not include mechanisms for farmer feedback, community interaction, or collaborative knowledge sharing.

2. Technology And Methodology

3.1 System Architecture

The Smart Crop Advisory System will be developed as a Java Spring Boot web application using microservices architecture for modularity and scalability.

3.1.1 Technology Stack

- **Backend:** Spring Boot, Spring Data JPA, Spring Security
- **Frontend:** HTML, CSS, JavaScript, AJAX
- **Database:** MySQL (for cloud storage and remote access)

3.1.2 AI & ML Components

- **Crop Recommendation Model:** Trained using historical soil and yield data with Machine Learning algorithms.
- **Fertilizer Optimization:** Regression and classification-based models for precise dosage calculation.
- **Plant Disease Detection:** Implemented using Deep Learning (CNN) for analyzing uploaded crop images.
- **Multilingual Chatbot:** NLP-based AI chatbot integrated with translation APIs and text-to-speech tools for voice support.

3.1.3 Notification System

- **WhatsApp Alerts:** Spring Boot backend triggers automated alerts for disease outbreaks and severe weather events.

3.1.4 Blog Page Implementation

- Developed using Spring Boot, JavaScript, and MySQL database.
- Supports CRUD operations and real-time updates using AJAX.

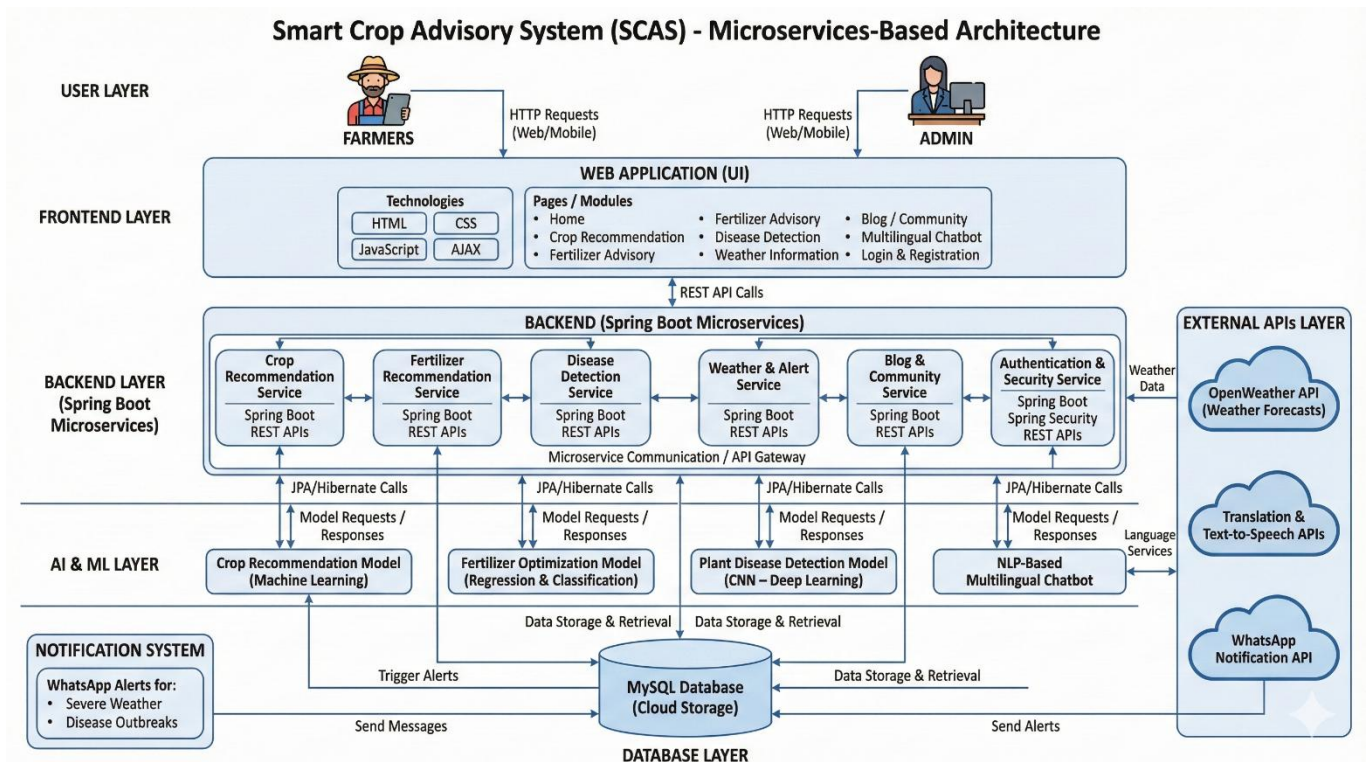


Figure 1: System Architecture Diagram for Smart Crop Advisory System (SCAS).

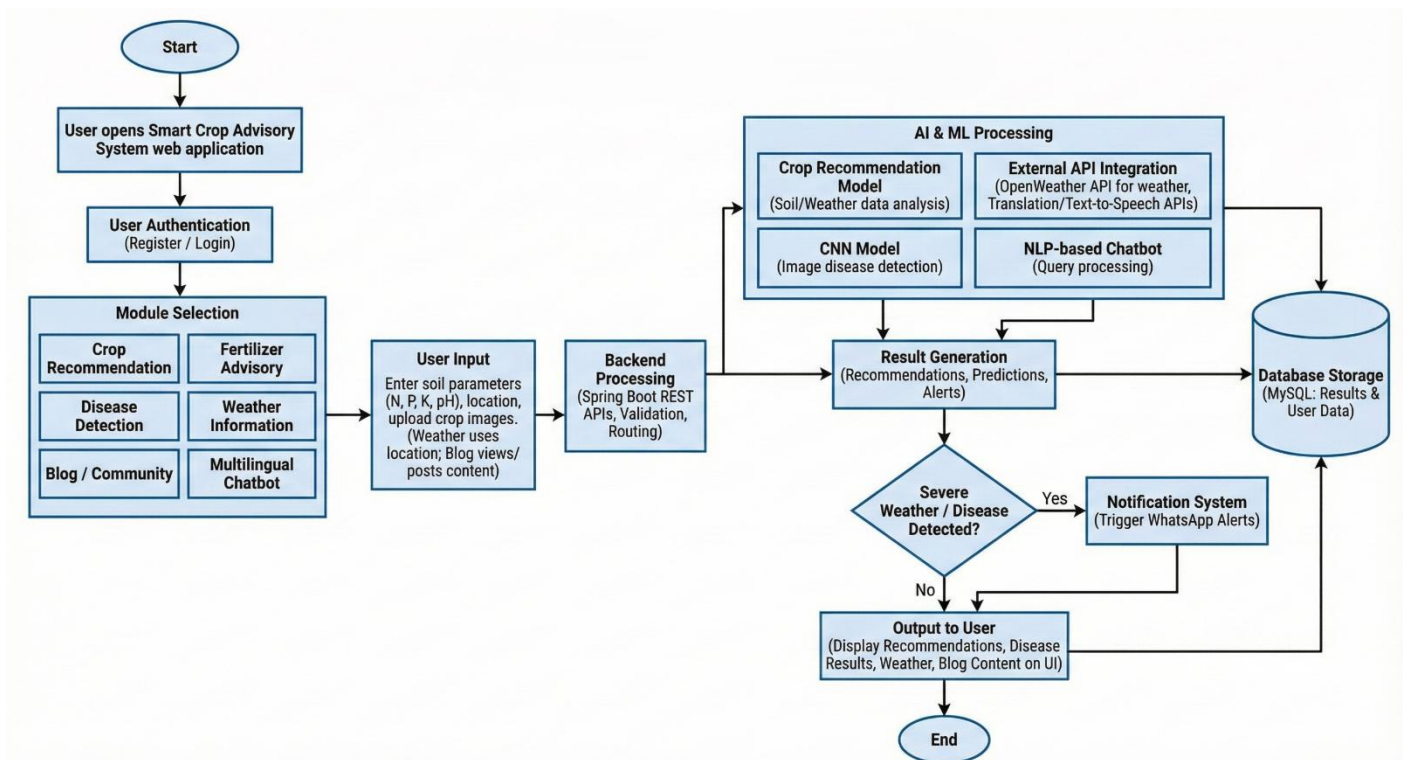


Figure 2: Workflow Diagram for Smart Crop Advisory System (SCAS).

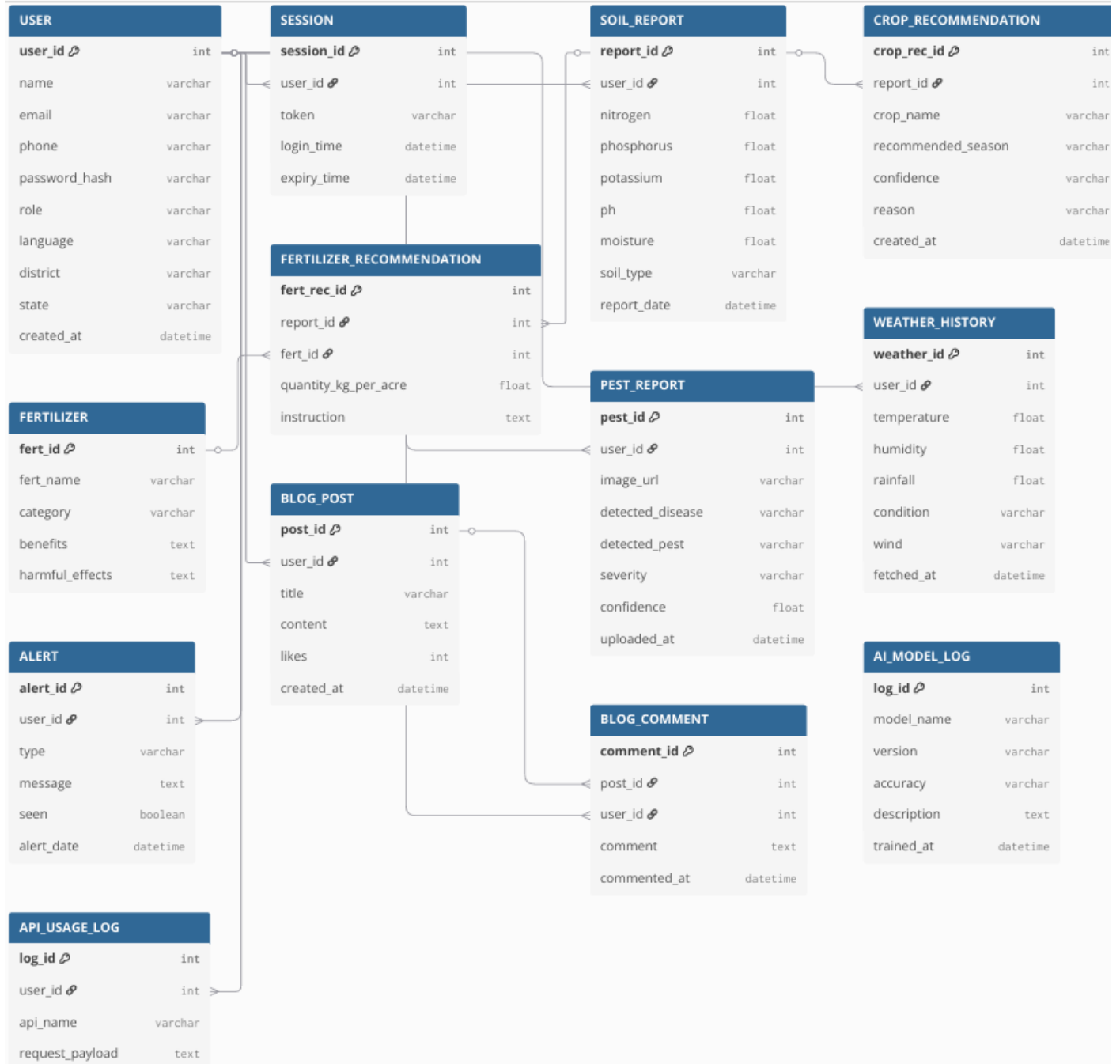


Figure 3: ER Diagram for Smart Crop Advisory System (SCAS).

3.2 Methodology

The methodology of the Smart Crop Advisory System (SCAS) is designed to provide intelligent, data-driven agricultural recommendations by integrating machine learning, deep learning, and web-based technologies. The process begins with the collection of agricultural datasets related to soil nutrients, crop types, fertilizer information, weather conditions, and disease images from reliable open-source platforms. The collected data is cleaned, normalized, and preprocessed to handle missing values and inconsistencies.

Machine learning algorithms are then trained using historical soil and weather data to develop an accurate crop recommendation model. Based on soil nutrient levels and selected crops, fertilizer recommendation models are implemented using classification and regression techniques to suggest optimal fertilizer types and usage. For crop protection, a deep learning-based Convolutional Neural Network (CNN) is used to detect diseases and weeds from uploaded crop images. Real-time weather data is integrated through the OpenWeather API to generate alerts for adverse climatic conditions. All components are integrated using a Spring Boot microservices architecture, ensuring modularity and scalability. The system uses MySQL for online data storage and SQLite for offline support, delivering a reliable, user-friendly, and scalable agricultural advisory platform.

3.3 Work Breakdown Structure (WBS)

Table 1: Work Breakdown Structure

S.No.	Phase	Task	Description
1	Planning	Problem study & requirement analysis	Identify key functional and non-functional requirements
2	Design	System architecture, database, and UI design	Create ERD, UML, and UI mockups
3	Development (Backend)	Spring Boot microservices	Implement modules for chatbot, alerts, and data analytics
4	Development (Frontend)	Responsive web UI	Build HTML/CSS/JS interface with localization support
5	AI Model Development	Data preprocessing, model training	Train ML models for crop/fertilizer/disease prediction
6	Integration	Connect backend, AI models, and APIs	Integrate OpenWeather, database, and chatbot
7	Testing	Unit, integration, and user testing	Validate accuracy, reliability, and usability
8	Documentation & Report	Final report, user manual	Prepare project report and demo documentation

3.4 Gantt Chart

Table 2: Gantt Chart

Week	Tasks	Details / Deliverables
Week 1	Requirement Analysis & System Design	Finalize project scope, prepare system architecture diagrams, and collect datasets related to soil, crops, fertilizers, and pests.
Week 2	Backend & Frontend Development	Develop Spring Boot backend services, configure MySQL database, and build a responsive frontend using HTML, CSS, and JavaScript.
Week 3	AI Integration & API Connectivity	Implement machine learning models for crop prediction and fertilizer recommendation, integrate CNN for disease detection, and connect OpenWeather API and chatbot services.
Week 4	Testing, Optimization & Deployment	Perform functional and user testing, fix bugs, integrate WhatsApp alert system, deploy the application, and prepare final documentation.

4 Results And Discussion

4.1 Expected Outcome

- The system is expected to provide **accurate crop recommendations** based on soil nutrients, weather conditions, helping farmers select suitable crops for improved yield.
- The fertilizer recommendation module will enable **optimized fertilizer usage**, reducing over-application, improving soil health, and lowering input costs.
- The disease and weed detection module is expected to support **identification of crop threats** using image-based deep learning models, thereby minimizing crop loss.
- Integration of real-time weather data will help generate **timely weather alerts**, allowing farmers to take preventive actions against adverse climatic conditions.
- The system is anticipated to **mitigate up to 60% of avoidable agricultural risks** related to crop failure, nutrient mismanagement, and disease outbreaks.
- Overall, SCAS will promote **data-driven and sustainable farming practices**, enhance productivity, and provide farmers with a reliable, user-friendly decision-support platform.

4.2 Final Outcome

The final outcome of the Smart Crop Advisory System (SCAS) is a fully integrated, AI-driven web platform that provides intelligent agricultural decision support to farmers. The system successfully combines crop recommendation, fertilizer advisory, disease and weed detection, and real-time weather alerts into a single, user-friendly application. By leveraging machine learning and deep learning techniques, SCAS delivers accurate and personalized recommendations based on soil parameters, environmental conditions, and crop images. The implementation of a Spring Boot microservices architecture ensures scalability, modularity, and efficient system performance. Additionally, the inclusion of a community and blog module promotes knowledge sharing and collaborative decision-making among farmers. Overall, the system demonstrates the practical application of Artificial Intelligence in agriculture, enabling farmers to reduce risks, optimize resources, improve crop productivity, and adopt sustainable farming practices through data-driven insights.

4.3 Future Scope

- **Multilingual and Voice-Based Support:** The system can be enhanced with multilingual and voice-enabled advisory features to make it accessible to farmers from different linguistic backgrounds.
- **IoT-Based Soil Monitoring:** Integration of IoT sensors can enable real-time monitoring of soil moisture, nutrient levels, and environmental conditions for more accurate recommendations.
- **Mobile Application Development:** A dedicated mobile application can be developed to improve accessibility and provide on-the-go advisory services to farmers.
- **Market Price Prediction and Supply Chain Integration:** Future versions may include market price forecasting and supply chain integration to help farmers make informed decisions regarding crop sales and distribution.

5 Conclusions

The Smart Crop Advisory System (SCAS) demonstrates the effective use of Artificial Intelligence and Machine Learning in addressing key challenges faced by modern agriculture. By integrating crop

recommendation, fertilizer advisory, disease and weed detection, and real-time weather alerts into a single platform, the system provides reliable and data-driven support to farmers. SCAS promotes sustainable farming practices, reduces agricultural risks, and enhances productivity, showcasing the potential of intelligent technologies in transforming traditional farming into precision agriculture.

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